CSCI 4150U Data Mining Cheat Sheet

Condensed from your uploaded review, clustering, anomaly detection, association analysis, SVM, and ensemble lecture PDFs. Letter landscape, 2 pages = one sheet double-sided.

1. Data Mining Basics

- **Data mining**: non-trivial extraction of implicit, previously unknown, and useful information from data.
- **Origins**: machine learning, pattern recognition, statistics, databases.
- **Main tasks**: classification, clustering, association rule mining, anomaly detection.

2. Data, Attributes, Quality

- **Object** = row / record. **Attribute** = column / feature.
- **Attribute types**: nominal, ordinal, interval, ratio.
- **Data forms**: record data, matrix data, transaction data, document data, graph data, ordered data.
- **Quality issues**: noise, outliers, duplicate data, missing values.

3. Similarity and Preprocessing

- **Distances / similarity**: Euclidean, Minkowski, Mahalanobis, cosine, simple matching, Jaccard, extended Jaccard.
- **Preprocessing**: aggregation, sampling, discretization, dimensionality reduction, feature subset selection.

4. Exploratory Analysis

- **Summary stats**: mean, standard deviation, frequency, mode, percentiles.
- **Visuals**: histogram, box plot, scatter plot, contour plot, data matrix / correlation matrix, parallel coordinates, star plots.
- **OLAP**: uses multidimensional array style representation for easier analysis than plain relational tables.

5. Classification Core

- **Goal**: learn a model that maps attributes x to predefined class label y.
- **Decision tree**: repeatedly split on attribute test conditions.
- **Best split**: compute impurity before split **P**, weighted impurity after split **M**, then maximize **Gain = P – M**.

6. Node Impurity

Gini = 1 – $\Sigma\, p_i^2$

Entropy = – $\Sigma\, p_i \log_2 p_i$

Misclassification error = 1 – $\max p_i$

Choose the split with the **lowest weighted child impurity** or highest gain.

7. Evaluation Metrics

Confusion matrix: TP, TN, FP, FN

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1 = 2PR / (P + R)

Cross-validation: split data into folds, test on one fold at a time, average performance.

Underfitting: model too simple, train and test errors high. **Overfitting**: train error low, test error high.

8. k-NN and Naive Bayes

- **k-NN**: compute distance from test point to training records, pick k nearest, predict by majority vote.
- **k-NN issues**: choice of k, distance metric, irrelevant features, scaling.
- **Naive Bayes**: assumes conditional independence of attributes given class.
- **Classify to class C_j** with largest $P(C_j)\ \Pi\ P(X_i|C_j)$.

9. SVM

Idea: find linear hyperplane that separates classes with **maximum margin**.

Decision boundary: $\mathbf{w}\cdot\mathbf{x} + b = 0$

Margins: $\mathbf{w}\cdot\mathbf{x} + b = +1$ and $\mathbf{w}\cdot\mathbf{x} + b = -1$

Margin width = 2 / $\|\mathbf{w}\|$

Hard-margin constraint: $y_i(\mathbf{w}\cdot\mathbf{x}_i + b) \geq 1$

Soft margin: add slack variables ξ_i and penalty C to allow violations.

Only **support vectors** determine the decision boundary.

10. Ensemble Methods

- **Ensemble**: train multiple classifiers and combine predictions.
- **General flow**: create multiple datasets → build multiple classifiers → combine outputs.
- **Bagging**: sample with replacement, train one base learner per bootstrap sample, combine by vote.
- Each instance has probability $1 - (1 - 1/n)^n$ of appearing at least once in a bootstrap sample.
- **Boosting**: adaptively increases weight of previously misclassified records and decreases weight of correctly classified ones.

11. Quick Exam Reminders

- Scale numeric features before distance-based methods.
- Tree splitting asks: which attribute reduces impurity most?
- High accuracy can mislead on imbalanced data, check precision, recall, F1.
- Cross-validation estimates generalization better than one random split.
- SVM margin bigger usually means better separation and robustness.

CSCI 4150U Data Mining Cheat Sheet, Side B

Clustering, anomaly detection, association rules, and high-yield reminders.

12. Clustering Basics

- **Clustering**: group similar points without labels.
- **Hard clustering**: each point belongs to one cluster only.
- **Soft / fuzzy clustering**: point can belong to multiple clusters with weights.

13. k-Means and Fuzzy c-Means

k-means objective: minimize SSE = $\Sigma_j \Sigma_{i \in C_j} \|x_i - c_j\|^2$ over assignments.
k-means loop: assign points to nearest centroid, recompute centroids, repeat.
Fuzzy c-means: weights $w_{ij} \in [0,1]$ and $\Sigma_j w_{ij}=1$.
Fuzzifier p > 1 controls how soft memberships are.

Update centroid: $c_j = \Sigma_i w_{ij}^p x_i / \Sigma_i w_{ij}^p$

Update weights: closer points get larger membership weights.

14. Hierarchical / CURE

- **Single link (MIN)**: can handle odd shapes but is sensitive to noise and outliers.
- **Complete link / group average / centroid**: may fail on non-globular clusters.
- **CURE**: uses multiple representative points per cluster, shrinks them toward the center by factor α, merges based on closest representative pair.
- CURE handles arbitrary shapes and sizes better, but cannot handle clusters with differing densities well.

15. Graph-Based and Spectral Clustering

- **Graph-based clustering**: build proximity graph, nodes = points, edge weights = similarity.
- **Sparsification**: keep only near-neighbor edges to reduce computation and noise.
- **Chameleon / Jarvis-Patrick / SNN**: graph or shared-neighbor based clustering families.
- **Spectral clustering**: use eigenvectors of similarity matrix to embed data in lower-dimensional space, then cluster there.
- Useful for **non-convex** cluster shapes such as two spirals where k-means fails.

16. GMM and EM

- **Gaussian mixture model**: data generated from a linear combination of K Gaussian components.
- Each point has a hidden **latent component** assignment.
- **Goal**: maximize log likelihood.
- **E-step**: compute responsibilities r_{ik} = probability point i belongs to component k.
- **M-step**: update π_k, μ_k, Σ_k using those responsibilities.
- Repeat E-step and M-step until log likelihood stops increasing.
- EM converges to a **local optimum**, so restart with different initializations.
- As covariance assumptions tighten, GMM behavior approaches k-means.

17. Anomaly Detection

- **Anomaly / outlier**: point considerably different from the rest; usually rare.
- **Causes**: different class, natural variation, data errors.
- **Noise ≠ anomaly**: noise is erroneous / contaminating and not necessarily interesting.
- **Challenges**: define normal region, unclear boundary, very few labels, evolving normal behavior.
- **Outputs**: binary label or anomaly score.

18. Anomaly Approaches

- **Visual**: scatter plots, box plots. Limitation: subjective, not automatic.
- **Statistical**: low probability under assumed distribution, examples include tests like Grubbs' test and likelihood methods.
- **Distance / proximity-based**: points far from others are outliers.
- **Density-based**: low-density points are outliers.
- **Cluster-based / model-based**: points that do not fit clusters or distort the model are anomalies.
- **Pattern matching**: compare to profiles of typical important events.

19. Association Rule Mining

Goal: find rules $X \Rightarrow Y$ that capture co-occurrence in transactions. Implication means **co-occurrence, not causality**.
Itemset: collection of items. **k-itemset**: has k items.
Support count $\sigma(X)$: number of transactions containing X.

Support $s(X) = \sigma(X) / |T|$

Confidence $c(X \Rightarrow Y) = s(X \cup Y) / s(X)$

Mining steps: 1) frequent itemset generation, 2) rule generation.
Apriori principle: if an itemset is frequent, all its subsets are frequent. Equivalently, if X is infrequent, all supersets of X are infrequent.

Anti-monotone support: if $X \subseteq Y$, then $s(X) \geq s(Y)$.

Apriori loop: generate candidates, prune, count support, eliminate infrequent candidates.
Rule count from itemset L of size k: $2^k - 2$ candidate rules.
Confidence pruning: for rules from same itemset, confidence is anti-monotone with respect to number of items on RHS.
Closed itemset: no superset with same support. **Maximal frequent itemset**: no frequent superset.